

Dixi Wang

phone: 425-499-1173 | email: wang6721@purdue.edu | LinkedIn: linkedin.com/in/dixi-wang

SUMMARY

Generative AI and machine learning researcher with a strong applied mathematics background in optimal transport, PDEs, optimization, and statistical learning theory. Experienced in scientific machine learning, generative modeling, and data-driven modeling. Seeking research scientist roles to apply theory-driven computational approaches to real-world AI problems.

PROFESSIONAL EXPERIENCE

Purdue University, West Lafayette, IN

Aug 2024 - present

Golomb Visiting Assistant Professor

- Conduct research in generative AI, transformer dynamics, optimal transport, and scientific machine learning.
- Teach undergraduate differential equations courses and mentor students in applied mathematics.

SKILLS

Programming Languages: Python, MATLAB, C++, SQL

Tech Stack and Tools: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Git, Jupyter, Linux

Languages: Chinese Mandarin (Native), English and Cantonese (Fluent), Spanish (Intermediate)

RESEARCH EXPERIENCE

Purdue University, West Lafayette, IN

Aug 2024 - present

Research Focus: Generative AI, Training Dynamics for Transformers, Optimal Transport, Learning Theory

- Developed a mean-field theory for Transformer soft-attention dynamics, establishing convergence results, adjoint-based optimality conditions, and supporting numerical experiments.
- Developed Transformer-based in-context learning methods for optimal transport, enabling few-shot distribution-to-distribution mapping without gradient updates at inference time.

University of Florida, Gainesville, FL

Sept 2019 - Jul 2024

Research Focus: Deep Learning, Operator Learning, Statistical Learning Theory, and PDEs

- Developed and implemented *Adaptive Friedrichs Learning*, a PyTorch-based deep learning algorithm for PDE-related problems; code available at github.com/dixiwan819/Adapted-Friedrich.
- Studied operator learning methods for Navier–Stokes equations and proved the existence of deep neural network approximations for solution operators.
- Designed and implemented PyTorch-based numerical experiments to validate the accuracy of neural operator learning algorithms.
- Collaborated on theoretical analysis of PDEs, including proving inviscid limit results for the Navier–Stokes equations.

EDUCATION

University of Florida, Gainesville, FL

Aug 2019 - Aug 2024

Ph.D. Mathematics: Partial Differential Equations and Deep Learning

The Chinese University of Hong Kong, Hong Kong, China

Aug 2017 - Jul 2019

M.S. Mathematics: Partial Differential Equations

The Chinese University of Hong Kong, Hong Kong, China

Sept 2013 - Jul 2017

B.S. Mathematics, Computational and Applied Mathematics Stream

Honors and Awards:

- Teaching Award, University of Florida Apr 2024
- Eleanor Ewing Ehrlich Award, University of Florida Jun 2022
- Undergraduate Full Scholarship Sept 2013

TEACHING EXPERIENCE

Purdue University, West Lafayette, IN Instructor: Ordinary Differential Equations, Differential Equations for Engineering students	Aug 2024 - present
University of Florida, Gainesville, FL Teaching Assistant: Calculus 1 and 2, Computational Linear Algebra (MATLAB)	Sept 2019 - May 2023
The Chinese University of Hong Kong, Hong Kong, China Instructor: Ordinary Differential Equations, Teaching Assistant: Linear Algebra	Aug 2017 - Jul 2019

PUBLICATIONS

- *Analyzing Training Dynamics of Transformers through Data-Driven Nonlocal Mean Field Control*, (with Y. Gao and R. Lai), preprint.
- *In-Context Operator Learning on the Space of Probability Measures*, (with F. Cole, Y. Chen, Y. Lu, and R. Lai), submitted.
- *Learning Operators for 3D Navier-Stokes Leray-Hopf Solutions by Neural Networks*, (with C. Yu), submitted.
- *Adaptive Friedrichs Learning: Weak Solutions of Partial Differential Equations via Deep Learning*, preprint.
- *Inviscid limit of the nonhomogeneous Navier-Stokes equations under Kolmogorov hypothesis*, (with C. Yu and X. Zhao), Dynamics of PDE, Vol.19, No.3, 191-206, 2022.

PRESENTATIONS

Applied Math Seminar, Georgia Institute of Technology	Apr 2026
SIAM New York-New Jersey-Pennsylvania Section Conference, Pennsylvania State University	Oct 2025
SIAM Central States Section Conference, University of Arkansas	Oct 2025
Partial Differential Equations Seminar, Purdue University	Sept 2024
Theory and Applications for Optimal Control and Generative Models, Purdue University	Aug 2024
Kinetic Seminar, The Chinese University of Hong Kong	Jun 2023
Workshop on Nonlinear PDEs: Stability Analysis and Multi-scale Applications, University of Pittsburgh	Apr 2023